

House Price Prediction: A Regression-Based Analysis of Residential Property Values

1. Project Overview

Business Context Residential property pricing is influenced by a combination of physical attributes (size, number of rooms, stories), amenities (air conditioning, hot water heating, parking), and locational advantages (main road access, preferred area). For real estate companies, investors, and buyers, understanding *which* of these factors drive price — and by how much — is essential for making sound pricing, investment, and purchasing decisions.

Objective This project builds and evaluates a regression-based model that predicts house prices from a set of structural and locational features, while also using statistical diagnostics to understand *why* the model behaves the way it does.

Why Predicting House Prices Matters Accurate price prediction reduces information asymmetry in the housing market. Sellers can avoid underpricing or overpricing a property, buyers can validate whether an asking price is fair, and real estate platforms can automate valuation estimates at scale.

Expected Business Value - A statistically grounded pricing benchmark that can support listing price recommendations. - Quantified evidence of which features (e.g., bathrooms, air conditioning, preferred area) most strongly influence price, supporting renovation and investment prioritization. - A transparent, interpretable model (as opposed to a black-box model) that stakeholders such as underwriters or appraisers can audit.

2. Business Problem

Property prices are the result of many interacting factors, and pricing a house without a systematic model is prone to guesswork and inconsistency. The business problem this project addresses is: **given a set of measurable house attributes, can we reliably estimate the price of a house, and can we identify which attributes matter most?** Solving this problem supports more consistent pricing, better investment targeting, and more informed negotiation for both buyers and sellers.

3. Project Goals

1. Build a regression model capable of predicting house price from structural and amenity-based features.
2. Quantify the statistical significance and business impact of each feature on price.

3. Detect and address multicollinearity among predictors.
 4. Test whether the model satisfies the core assumptions of linear regression (linearity, homoscedasticity, normality of residuals, absence of autocorrelation).
 5. Improve model diagnostics through transformation (log-price) where the original model falls short.
 6. Translate statistical findings into actionable business insights and recommendations for real estate stakeholders.
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4. Dataset Description

- **Source:** Local file `Housing.csv` (a standard housing valuation dataset)
- **Observations:** 545 rows
- **Variables:** 13 columns (1 target + 12 features)
- **Target Variable:** price
- **Missing Values:** None (`df.isnull().sum()` returned 0 for every column)
- **Duplicate Records:** None (`df.duplicated().sum()` returned 0)

Data Dictionary

Column	Description	Original Data Type	Role
price	Sale price of the house	int64	Target variable
area	Total plot/floor area of the house	int64	Numerical feature
bedrooms	Number of bedrooms	int64	Numerical feature
bathrooms	Number of bathrooms	int64	Numerical feature
stories	Number of stories (floors) in the house	int64	Numerical feature
mainroad	Whether the house is directly connected to a main road (yes/no)	object	Binary categorical
guestroom	Whether the house has a guest room (yes/no)	object	Binary categorical
basement	Whether the house has a basement (yes/no)	object	Binary categorical
hotwaterheating	Whether the house has a hot water heating system	object	Binary categorical

Column	Description	Original Data Type	Role
airconditioning	(yes/no) Whether the house has air conditioning	object	Binary categorical
parking	(yes/no) Number of parking spaces available	int64	Numerical feature
prefarea	Whether the house is located in a preferred/desirable area (yes/no)	object	Binary categorical
furnishingstatus	Furnishing level of the house (furnished / semi-furnished / unfurnished)	object	Multi-class categorical

Descriptive statistics (numerical columns, from `df.describe()`):

Feature	Mean	Std Dev	Min	25%	50%	75%	Max
price	4,766,729	1,870,440	1,750,000	3,430,000	4,340,000	5,740,000	13,300,000
area	5,150.5	2,170.1	1,650	3,600	4,600	6,360	16,200
bedrooms	2.97	0.74	1	2	3	3	6
bathrooms	1.29	0.50	1	1	1	2	4
stories	1.81	0.87	1	1	2	2	4
parking	0.69	0.86	0	0	0	1	3

5. Project Workflow

Business Understanding



Data Collection



Data Cleaning



Exploratory Data Analysis (EDA)



Feature Engineering



Correlation & Multicollinearity Analysis



Regression Modeling (Linear Regression)



Model Evaluation (MAE, RMSE, R^2)



Statistical Modeling (OLS) & Model Refinement



Regression Assumption Testing (Residuals, Q-Q, Breusch-Pagan, Durbin-Watson)



Log Transformation & Model Comparison



Business Insights & Recommendations

6. Data Cleaning

Step	Why It Was Needed	What Was Done	Business Impact
Missing value check	Missing data can bias coefficient estimates and break model training	Ran <code>df.isnull().sum()</code> across all columns; confirmed zero missing values	No imputation was required, preserving the full 545-record sample and avoiding artificial bias
Duplicate record check	Duplicate rows can artificially inflate the influence of certain data points	Ran <code>df.duplicated().sum()</code> ; confirmed zero duplicates	Ensures every price observation reflects an independent property, keeping the model's learned relationships representative
Data type review	Regression algorithms require numeric input; the raw dataset mixes numeric and text (object) columns	Reviewed <code>df.info()</code> ; identified 6 numeric columns and 7 object columns	Clarified which columns needed encoding before modeling
Encoding preparation	Categorical yes/no and furnishing-status text values cannot be fed directly into a regression model	Addressed in the Feature Engineering stage (binary mapping and one-hot encoding)	Enables the model to quantify the numeric effect of qualitative attributes like air conditioning or furnishing on price

7. Exploratory Data Analysis

7.1 Categorical Feature Distributions (Count Plots)

Purpose: Understand how common or rare each amenity/attribute is across the 545 properties, since rare categories can be less reliable predictors.

Key Observations: - The majority of homes have `mainroad = yes`, with only a small minority off the main road. - Most homes do **not** have a `guestroom`, `basement`, or `hotwaterheating` — these are minority amenities. - `airconditioning = no` is more common than `airconditioning = yes`. - Most homes are **not** in a `prefarea`. - `furnishingstatus` is fairly spread across the three categories, with “semi-furnished” being the most common, followed by “unfurnished,” then “furnished.” - Among numeric-but-discrete features, most homes have 2–3 bedrooms, 1 bathroom, 1–2 stories, and 0 parking spaces.

Business Insight: Because amenities like hot water heating, a guest room, and a basement are relatively rare, homes that *do* have them are differentiated from the bulk of the market — this scarcity is consistent with these features later showing a meaningful positive effect on price.

7.2 Area Distribution (Histogram)

Purpose: Assess the shape of the plot-area distribution to check for skew that could affect linear modeling.

Key Observation: The area distribution is right-skewed, concentrated between roughly 3,000–6,500 sq. ft., with a long tail extending toward 16,200 sq. ft.

Business Insight: A small number of very large properties exist in the market; these will likely act as high-leverage points in a linear regression and should be monitored during diagnostics.

7.3 Area vs. Price (Scatter Plot)

Purpose: Visually inspect the relationship between plot area and price, since area is expected to be a primary price driver.

Key Observations (from the notebook’s own annotated insights): - Many houses are clustered in the cheap-to-moderate price range. - Beyond that cluster, prices increase rapidly, reflecting the combined influence of multiple features beyond area alone.

Business Insight: Area alone does not fully explain price — the widening spread at higher area values signals that other features (bathrooms, air conditioning, location) interact with area to determine final price, justifying a multivariate regression rather than a simple area-only model.

7.4 Price Distribution (Histogram)

Purpose: Examine the shape of the target variable, since linear regression performs best when the target is not heavily skewed.

Key Observation: price is right-skewed, with most properties priced between roughly 2–6 million and a long tail of high-value properties extending to 13.3 million.

Business Insight: This skew is the direct motivation for the later log-transformation experiment — a right-skewed target commonly violates the normality-of-residuals assumption in linear regression.

7.5 Price and Area Boxplots (Outlier Verification)

Purpose: Formally check for outliers in both the target (price) and a key predictor (area).

Key Observation: Both boxplots show multiple points beyond the upper whisker, confirming the presence of high-value outliers in price and large-area outliers in the feature space.

Business Insight: A small set of premium properties exist that are meaningfully more expensive and larger than the rest of the market — these outliers were later formally quantified (Section 15) and explicitly retained rather than removed.

7.6 Correlation Heatmap

Purpose: Quantify linear relationships between all numeric/encoded features and price to identify likely price drivers before modeling.

Key Observations (as annotated in the notebook): - Price shows a strong positive correlation with area (0.54), bathrooms (0.52), and airconditioning (0.45). - Price shows a noticeable **negative** correlation with basement (-0.19). - area is moderately linked with parking (0.35). - No strong negative correlations exist elsewhere in the dataset; most relationships are weak or positively inclined.

Business Insight: Area, bathrooms, and air conditioning stand out early as the strongest linear price drivers, giving a preliminary signal of which features the regression model is likely to weight most heavily — though the counterintuitive negative basement correlation warranted closer inspection once other variables were controlled for in the multivariate model (where basement in fact showed a **positive**, statistically significant coefficient — see Section 12), illustrating why bivariate correlation should not be read as a standalone causal signal.

8. Feature Engineering

Binary Encoding The six yes/no categorical columns — mainroad, guestroom, basement, hotwaterheating, airconditioning, and prefarea — were mapped to 1

(yes) and 0 (no). This was necessary because linear regression requires strictly numeric inputs; binary mapping preserves the full information content of these two-level variables without adding unnecessary dimensionality.

Dummy Variable Creation The `furnishingstatus` column (3 categories: furnished, semi-furnished, unfurnished) was one-hot encoded using `pd.get_dummies(..., columns=["furnishingstatus"], drop_first=True)`, producing two new columns: `furnishingstatus_semi-furnished` and `furnishingstatus_unfurnished`.

Reference Category Because `drop_first=True` was used, furnished became the implicit **reference category**. All coefficients on the two dummy columns are therefore interpreted *relative to a fully furnished home*.

Why `drop_first=True` Was Used Including a dummy variable for all three furnishing categories alongside an intercept term would create perfect multicollinearity (the “dummy variable trap”), since the three dummies would sum to a constant. Dropping one category avoids this redundancy while losing no information — the dropped category’s effect is captured by the model’s intercept and is recoverable by comparison with the other two dummy coefficients.

9. Correlation Analysis

Purpose: To surface likely linear predictors of price prior to committing to a full multivariate model, and to sanity-check the direction of relationships.

Findings: As detailed in Section 7.6, area (0.54), bathrooms (0.52), and airconditioning (0.45) were the strongest positive correlates of price, while basement showed a mild negative correlation (-0.19) at the bivariate level.

Business Interpretation: Larger homes, homes with more bathrooms, and homes with air conditioning command a price premium in raw, single-variable terms. However, correlation analysis alone cannot separate the individual contribution of each feature when features are themselves correlated with one another (e.g., larger homes tend to have more bathrooms) — this limitation is precisely what the subsequent multicollinearity and multivariate regression analysis were designed to address.

10. Multicollinearity Analysis

Why VIF Is Important Variance Inflation Factor (VIF) measures how much a predictor’s variance is inflated due to its linear correlation with other predictors. High multicollinearity does not bias predictions, but it destabilizes coefficient estimates and their statistical significance, making it harder to isolate the individual effect of each feature — a key requirement for this project’s goal of identifying which attributes matter most.

VIF Results (from the notebook):

Feature	VIF
bedrooms	16.65
bathrooms	9.42
area	8.28
stories	7.88
mainroad	6.88
furnishingstatus_semi-furnished	2.39
basement	2.02
furnishingstatus_unfurnished	2.01
parking	1.99
airconditioning	1.77
prefarea	1.49
guestroom	1.47
hotwaterheating	1.09

Interpretation - bedrooms (VIF = 16.65) shows **severe** multicollinearity — well above the common threshold of 10. - bathrooms (9.42), area (8.28), stories (7.88), and mainroad (6.88) show **moderate** multicollinearity (VIF between 5 and 10). - All remaining features have VIF below 5, indicating acceptable, low multicollinearity.

Why bedrooms Had a High VIF As noted in the notebook, the number of bedrooms is closely tied to overall house size — larger homes (higher area) tend to have more bedrooms and bathrooms. This overlap means bedrooms carries little unique information once area and bathrooms are already in the model, producing its inflated VIF.

Why Variables Were Not Immediately Removed Rather than dropping high-VIF variables purely on the basis of VIF, the notebook's approach was to first fit the full model and examine each variable's statistical significance in the OLS results. This is a more rigorous approach because a high VIF flags *potential* redundancy but does not, by itself, prove a variable is uninformative — that judgment was reserved for the p-value evidence gathered in the OLS regression (Section 12), where bedrooms was confirmed to be statistically insignificant before it was removed in the refined model (Section 13).

11. Linear Regression

Why Linear Regression Was Chosen Linear regression was selected as the baseline model because it is highly interpretable — each coefficient directly quantifies the price impact of a one-unit change in a feature — which aligns with the project's goal of explaining *why* prices move, not just predicting a number. It is also a natural fit given the dataset's continuous target and mix of continuous/binary predictors.

Model Assumptions Classical linear regression relies on: (1) a linear relationship between predictors and the target, (2) independence of residuals, (3) homoscedasticity (constant residual variance), (4) normally distributed residuals, and (5) limited multicollinearity among predictors. These assumptions were explicitly tested later in the project (Section 14).

Train-Test Split The data was split 80/20 into training and test sets using `train_test_split(X, y, test_size=0.2, random_state=42)`, holding out 20% of observations to evaluate the model on unseen data and reduce the risk of reporting an overly optimistic, overfit performance figure.

Training Process A `sklearn.linear_model.LinearRegression` model was fit on the training features (with all 13 encoded predictors, including bedrooms) and the raw price target.

12. Model Evaluation (Initial Full Model)

The initial model was evaluated on the held-out test set using three standard regression metrics:

Metric	Value	Interpretation
MAE	970,043.40	On average, the model's price predictions are off by approximately 970,043 currency units in absolute terms
MSE	1,754,318,687,330.66	The squared-error metric used internally to compute RMSE; not directly interpretable in price units
RMSE	1,324,506.96	The typical prediction error, in the same units as price, penalizing larger errors more heavily than MAE
R ²	0.6529	The model explains approximately 65.3% of the variance in house prices on the test set

Business Interpretation: An R² of roughly 65% indicates the model captures a substantial share of what drives price, but a typical error of over 1.3 million (RMSE) relative to a median price of 4.34 million means predictions should be treated as a *directional estimate*, not a precise appraisal, at this stage of the analysis.

13. Statistical Model (OLS) — Full Feature Set

An Ordinary Least Squares (OLS) regression was fit using `statsmodels` on the full dataset (not just the training split) to obtain interpretable coefficients, standard errors, and significance tests.

Overall Model Fit - R^2 : 0.682 — the full model explains 68.2% of the variance in price. - **Adjusted R^2 :** 0.674 — after penalizing for the number of predictors, the explanatory power remains high, indicating the model is not simply overfitting with extra variables. - **F-statistic:** 87.52, with **Prob (F-statistic) = $9.07e-123$** — an extremely small p-value confirming that the model as a whole is statistically significant; the predictors jointly explain price far better than chance.

Coefficients, Significance, and Business Meaning

Feature	Coefficient	p-value	Significant?	Business Meaning
const	4.28e+04	0.872	No	Baseline intercept, not statistically distinguishable from zero
area	244.14	0.000	Yes	Each additional unit of area adds ~244 to price, holding other features constant
bedrooms	1.148e+05	0.114	No	No statistically reliable effect once other size-related variables are controlled for
bathrooms	9.877e+05	0.000	Yes	Each additional bathroom adds ~988,000 to price
stories	4.508e+05	0.000	Yes	Each additional story adds ~451,000 to price
mainroad	4.213e+05	0.003	Yes	Main-road access adds ~421,000 to

Feature	Coefficient	p-value	Significant?	Business Meaning
				price
guestroom	3.005e+05	0.023	Yes	Having a guest room adds ~301,000 to price
basement	3.501e+05	0.002	Yes	Having a basement adds ~350,000 to price
hotwaterheating	8.554e+05	0.000	Yes	Hot water heating adds ~855,000 to price
airconditioning	8.650e+05	0.000	Yes	Air conditioning adds ~865,000 to price
parking	2.771e+05	0.000	Yes	Each additional parking space adds ~277,000 to price
prefarea	6.515e+05	0.000	Yes	Being in a preferred area adds ~652,000 to price
furnishingstatus_semi-furnished	-4.634e+04	0.691	No	No statistically reliable difference from a fully furnished home
furnishingstatus_unfurnished	-4.112e+05	0.001	Yes	An unfurnished home sells for ~411,000 less than a comparable furnished home

Confidence Intervals: All statistically significant coefficients have 95% confidence intervals that exclude zero (e.g., bathrooms: 785,000 to 1,190,000; airconditioning: 652,000 to 1,080,000), reinforcing that these effects are unlikely to be due to chance.

Why bedrooms and furnishingstatus_semi-furnished Were Removed Both variables failed to reach conventional statistical significance ($p = 0.114$ and $p = 0.691$, respectively), meaning the data does not provide strong evidence that they have a real, independent effect on price once the other correlated features are accounted for. For bedrooms, this is consistent with the high VIF (16.65) identified earlier — its apparent influence on price is likely already captured by area and bathrooms. For furnishingstatus_semi-furnished, the effect is simply too close to zero and too imprecisely estimated to be distinguished from “no difference vs. a furnished home.”

Model Assumption Concerns Noted at This Stage The notebook flags that (1) the condition number (3.49×10^4) is large, suggesting multicollinearity or numerical instability, and (2) the Omnibus (97.909) and Jarque-Bera (258.281) tests both reject normality of residuals ($p < 0.001$), meaning the classical OLS assumptions are not fully satisfied by this model — motivating the diagnostic testing and log-transformation work that follows.

14. Model Refinement

Why Insignificant Variables Were Removed bedrooms and furnishingstatus_semi-furnished were dropped from the feature set because they were not statistically significant in the full OLS model (Section 13) and their presence was linked to multicollinearity (in the case of bedrooms). Removing statistically uninformative predictors produces a more parsimonious, more interpretable model without meaningfully sacrificing explanatory power.

Comparing the Two Models

Metric	Full Model (sklearn)	Refined Model (sklearn)
MAE	970,043.40	965,607.88
RMSE	1,324,506.96	1,324,523.22
R ²	0.6529	0.6529

Why the Refined Model Is Better The refined model achieves virtually identical predictive performance (R^2 unchanged at 0.6529, RMSE effectively unchanged, MAE marginally improved) while using two fewer predictors. This confirms that bedrooms and furnishingstatus_semi-furnished were not contributing meaningful predictive information — the refined model is preferable on the grounds of parsimony and interpretability: every remaining coefficient can be trusted to reflect a real, independent relationship with price, rather than noise from redundant variables.

15. Regression Assumption Testing (Refined Model)

15.1 Residual Plot

The residual plot (predicted values vs. residuals) shows residuals scattered around the zero line without an obvious curved pattern, which is consistent with the linearity assumption being reasonably well satisfied. However, the spread of residuals is not perfectly uniform across the range of predicted values, hinting at the heteroscedasticity that the Breusch-Pagan test later confirms.

Assumption status: Linearity — reasonably satisfied; homoscedasticity — questionable (see 15.3).

15.2 Q-Q Plot

The Q-Q plot of residuals shows the points following the reference line closely in the middle of the distribution but diverging at both tails, particularly the upper tail. This indicates the residuals have fatter tails than a normal distribution — i.e., more extreme prediction errors than a normal-error model would expect.

Assumption status: Normality of residuals — not fully satisfied, especially at the tails.

15.3 Breusch-Pagan Test

Results: LM Statistic = 68.416, LM p-value = 1.57e-09; F Statistic = 5.864, F p-value = 3.49e-10.

Since both p-values are far below 0.05, the null hypothesis of homoscedasticity (constant residual variance) is rejected.

Assumption status: Homoscedasticity — violated (heteroscedasticity is present).

15.4 Durbin-Watson Test

Result: Durbin-Watson statistic = 1.209.

A value of 2.0 indicates no autocorrelation; values below 2 suggest positive autocorrelation. At 1.209, the residuals show **mild positive autocorrelation**.

Assumption status: Independence of residuals — mildly violated.

15.5 Summary of Diagnostics

The refined linear model captures the linear structure of the data reasonably well and is statistically significant overall, but it does not fully satisfy the classical assumptions of normal, homoscedastic, independent residuals. This is the direct motivation for testing a log-transformed version of the target variable in the next section.

16. Log Transformation

Why log(price) Was Introduced The price distribution is right-skewed (Section 7.4), and the diagnostic tests in Section 15 confirmed non-normal, heteroscedastic residuals. A logarithmic transformation of the target is a standard remedy for right-skewed monetary targets: it compresses the influence of high-value outliers and tends to stabilize variance, since percentage-based price differences (rather than absolute differences) become more linear in log space.

How It Improved the Model After transforming, `log_price` was modeled against the same refined feature set (excluding bedrooms and furnishingstatus_semi-furnished).

Model Comparison: Original (Full), Refined, and Log-Transformed

Aspect	Original Full Model	Refined Model	Log-Transformed Model
Target	price	price	log(price)
R ² (OLS)	0.682	<i>(not separately reported via OLS; sklearn R² = 0.6529)</i>	0.696
Adjusted R ² (OLS)	0.674	<i>(not separately reported)</i>	0.690
MAE (sklearn)	970,043.40	965,607.88	0.2014 (log scale)
RMSE (sklearn)	1,324,506.96	1,324,523.22	0.2535 (log scale)
R ² (sklearn, test set)	0.6529	0.6529	0.6671
Durbin-Watson	1.209	1.209	1.292
Breusch-Pagan p-value	3.49e-10	3.49e-10	0.0073
Omnibus p-value	0.000	0.000	0.178

Note: MAE and RMSE for the log-transformed model are expressed in log-price units and are not directly comparable in magnitude to the original models' currency-unit errors; they are included for completeness as reported in the notebook.

Improvements Observed - R² and Adjusted R²: Both the sklearn test-set R² (0.6529 → 0.6671) and the OLS R²/Adjusted R² (0.682/0.674 → 0.696/0.690) improved after the log transformation. - **Residual Normality:** The Omnibus test p-value rose from 0.000 (full model) to 0.178 (log model) — since 0.178 > 0.05, the log model's residuals **fail to reject** the null hypothesis of normality, a marked improvement. This is visually supported by the log model's Q-Q plot, whose points track the reference line more closely than the original model's Q-Q plot. - **Heteroscedasticity:** The Breusch-Pagan p-value increased from 3.49e-10 to 0.0073. While still below 0.05 — meaning heteroscedasticity is not fully eliminated — the severity is substantially reduced, as reflected in the much smaller LM Statistic (68.42 → 25.41) and F Statistic (5.86 → 2.37). - **Overall Model Quality:** The log-transformed model

shows better-behaved residuals, closer-to-normal errors, and improved explanatory power, making it the preferred model in this analysis, consistent with the notebook's own conclusion.

17. Model Comparison Table

Criterion	Original (Full) Model	Refined Model	Log-Transformed Model
Features Used	All 13 (incl. bedrooms, both furnishing dummies)	11 (excl. bedrooms, furnishingstatus_se mi-furnished)	11 (same as Refined, target log-transformed)
R² (sklearn test set)	0.6529	0.6529	0.6671
R² (OLS, full data)	0.682	—	0.696
Adjusted R² (OLS)	0.674	—	0.690
MAE	970,043.40	965,607.88	0.2014 (log scale)
RMSE	1,324,506.96	1,324,523.22	0.2535 (log scale)
Durbin-Watson	1.209	1.209	1.292
Breusch-Pagan (heteroscedasticity)	Violated (p = 3.49e-10)	Violated (p = 3.49e-10)	Reduced severity (p = 0.0073)
Residual Normality (Omnibus)	Violated (p = 0.000)	Violated (p = 0.000)	Not rejected (p = 0.178)
Advantages	Simplest to interpret in raw currency units	Removes noisy, insignificant predictors; more parsimonious	Best statistical properties; coefficients interpretable as approximate % price change
Limitations	Includes statistically insignificant variables; weaker diagnostics	Same diagnostic issues as original model	Coefficients require conversion to interpret in currency terms; heteroscedasticity not fully eliminated

18. Business Insights

1. **Bathrooms carry outsized price weight.** With a full-model coefficient of ~988,000 and the second-highest raw correlation with price (0.52), each additional bathroom is one of the single largest levers on valuation.
2. **Air conditioning is a high-value amenity.** A coefficient of ~865,000 (full model) and ~0.173 in log terms (roughly consistent with a large percentage price premium) makes air conditioning one of the strongest amenity-based price drivers.
3. **Stories add substantial value.** Each additional story is associated with a ~451,000 increase in price (full model), reflecting a market premium for multi-level homes.
4. **Parking is a statistically significant, moderate driver.** Each additional parking space adds ~277,000 to price (full model) and remains significant in the log model (0.046, $p < 0.001$).
5. **Preferred-area location commands a strong premium.** *prefarea* adds ~652,000 in the full model and shows a large, significant log-scale coefficient (0.127) — location remains a top-tier price factor even after controlling for structural attributes.
6. **A basement adds value once other factors are controlled for.** Despite a mildly negative *bivariate* correlation with price (-0.19), the multivariate model shows a positive, statistically significant basement effect (~350,000, full model), illustrating that basements co-occur with other lower-priced home characteristics in the raw data, but independently increase price when isolated.
7. **Main-road access matters.** A ~421,000 premium (full model) confirms that accessibility is capitalized into price.
8. **Furnishing status has an asymmetric effect.** Being unfurnished is associated with a clear, significant price discount (~411,000 in the full model; -0.124 in log terms), while being semi-furnished shows no statistically distinguishable difference from being fully furnished.
9. **Hot water heating, though rare in the dataset, carries a large premium.** Its full-model coefficient (~855,000) is among the largest of any feature, suggesting this uncommon amenity is highly valued where present.
10. **Area has a modest but highly significant per-unit effect.** At ~244 per unit of area (full model) and consistently significant across every model version, area functions as a steady baseline driver rather than the dominant one.
11. **Guest rooms provide a real but comparatively smaller premium** (~301,000, full model; 0.068 log-scale), positioning this amenity as a secondary rather than primary value driver.
12. **Bedrooms do not independently predict price** once area and bathrooms are included — a counter-intuitive but statistically well-supported finding with direct implications for renovation decisions (see Recommendations).
13. **A small set of premium properties (15 outliers by IQR) exist at the top of the market,** and their retention (rather than removal) in the log-transformed model reflects a deliberate choice to preserve real market signal rather than discard legitimate high-value listings.

14. **The log-transformed model's improved normality and reduced heteroscedasticity indicate that percentage-based pricing effects (e.g., "+15% for air conditioning") are a more statistically defensible way to communicate these findings than flat currency estimates.**
 15. **No single feature dominates price formation** — the model's ~65–70% explained variance indicates a meaningful but incomplete picture, implying that roughly 30–35% of price variation stems from factors outside this dataset (e.g., neighborhood-level dynamics, market timing, property condition).
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19. Recommendations

For Real Estate Companies - Use the log-transformed model's coefficients to communicate approximate percentage price premiums (e.g., for air conditioning, preferred area) to sellers and buyers, since this framing is more statistically sound than the raw-currency full model. - Prioritize verifying bathroom count, air conditioning, and preferred-area status during listing intake, as these are the most consistently significant value drivers across all model versions.

For Property Investors - Focus renovation or acquisition strategy on high-impact, statistically validated features — additional bathrooms, air conditioning, and parking — rather than adding bedrooms, which showed no independent price effect in this dataset. - Treat basement-equipped properties as undervalued opportunities if their basement is not being reflected in comparable pricing, since the controlled (multivariate) effect of a basement is positive despite a misleading raw negative correlation.

For Home Buyers - Use the model's significant features as a negotiation checklist: confirm whether a listing's price is justified by its area, bathrooms, stories, parking, air conditioning, and location, rather than by bedroom count alone. - Recognize that an unfurnished home carries a statistically supported discount, which may present value opportunities for buyers willing to furnish independently.

For Developers - Given that bedrooms showed no independent significance, consider whether reallocating square footage toward additional bathrooms or amenities (air conditioning, hot water heating) yields a better return than maximizing bedroom count.

Pricing Strategy - Adopt the log-transformed model as the primary reference model for pricing guidance, given its superior statistical diagnostics (better-behaved residuals, closer-to-normal errors), while treating its output as a percentage-adjustment framework rather than a single-point currency prediction.

20. Limitations

Data Limitations - The dataset does not include neighborhood/location identifiers beyond the binary prefarea and mainroad flags, limiting the model's ability to capture fine-grained locational effects. - No time dimension (e.g., listing date, sale date) is present, so the

model cannot account for market trends or seasonality. - The dataset size (545 records) is modest, which constrains the granularity of subgroup analysis.

Model Assumption Limitations - Diagnostic testing confirmed that even the best-performing (log-transformed) model still shows statistically significant heteroscedasticity (Breusch-Pagan $p = 0.0073$) and mild positive residual autocorrelation (Durbin-Watson = 1.292), meaning standard errors and confidence intervals should be interpreted with some caution. - Multicollinearity, while reduced by removing bedrooms, is still present at a structural level (e.g., area, bathrooms, and stories remain correlated), and the condition number in both OLS models remains large.

External Factors Not Captured - Property condition, age, renovation history, school district quality, crime rates, and macroeconomic conditions (interest rates, local demand cycles) are not present in the dataset but plausibly influence real-world price.

21. Future Improvements

- **Robust Regression:** Apply robust regression techniques (e.g., Huber or RANSAC regressors) to reduce the influence of the 15 identified price outliers without discarding them outright.
 - **Random Forest / Gradient Boosting / XGBoost:** Explore tree-based ensemble models to capture potential non-linear relationships and feature interactions that linear regression cannot represent.
 - **Feature Engineering:** Introduce interaction terms (e.g., area $\times$ bathrooms) or polynomial features to better capture the non-uniform price increase observed in the area-vs-price scatter plot.
 - **Cross-Validation:** Replace the single train-test split with k-fold cross-validation to obtain more stable and generalizable performance estimates.
 - **Hyperparameter Tuning:** For any future tree-based or regularized linear models (e.g., Ridge, Lasso), apply systematic hyperparameter search to optimize the bias-variance tradeoff.
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22. Conclusion

This project developed and rigorously evaluated a linear regression framework for predicting house prices from structural, amenity, and locational features. Starting from a clean, complete dataset of 545 properties, the analysis progressed through exploratory analysis, multicollinearity diagnostics, a full multivariate OLS model, a refined model with statistically insignificant predictors removed, and finally a log-transformed model that improved both predictive performance (R^2 of 0.696 vs. 0.682) and statistical validity (residual normality no longer rejected, heteroscedasticity substantially reduced). Bathrooms, air conditioning, preferred-area location, and stories emerged as the most consistent and economically meaningful drivers of price, while bedroom count showed no independent effect once home size was already accounted for. The log-transformed model,

despite retaining some residual heteroscedasticity and autocorrelation, is recommended as the most statistically defensible model produced in this analysis, and the documented limitations and future improvement paths provide a clear roadmap for extending this work toward production-grade valuation modeling.

Documentation generated from direct analysis of the accompanying Jupyter Notebook. All figures, coefficients, and statistical results cited above are sourced exclusively from the notebook's executed outputs.